Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/14/2005
PI Name:	Losert, Wolfgang Ph.D.		
Project Title:	Granular Flow Instabilities: Transients, Aging, and Segregation Dynamics		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-3511	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	01/01/2002	**End Date:**	12/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Friedmann, Julio (University of Maryland)		
Grant/Contract No.:	NAG3-2736		
Performance Goal No.:			
Performance Goal Text:			

The aim of this project is an interdisciplinary study of granular flow instabilities, involving both laboratory experiments for the development of physical models, as well as approaches and data sets from geology. The central aim of the proposed work is the improvement of physical models for granular flow instabilities, and the application of those models to rock avalanches. The proposed research will provide experimental data and test theoretical models to address open questions about three main aspects of granular flow instabilities:
- Flow initiation and cessation: Granular matter can exhibit a sudden transition from a solid-like state to a flowing state and vice versa, e.g. during avalanching and jamming. We will investigate particle motions during such transients. One question is, what timescales and spatial scales best describe such a transition?

Task Description:	- Dynamics of aging: Even an apparently stationary granular assembly slowly rearranges (aging). This causes changes in material properties, such as the gradual increase in yield strength. We will measure, the changes in the network of particle contacts directly with confocal imaging. - Dynamics of segregation: Grains of different size or shape will segregate under some conditions in a driven granular material. While models for segregation exist, several fundamental issues warrant further study. We will relate segregation to the sharing of momentum and energy between particle species. Understanding granular flow instabilities is of fundamental importance since it may reveal the most suitable equations to describe granular flows. Flow instabilities are also of great technological significance for a broad range of industries. The sudden jamming of grains in confined geometries, e.g. in a hopper, is a significant problem for food, mining, and raw materials processing industries. The slow creep and gradual strengthening of a quasi-stationary granular pile under stress can lead to structural damage of storage containers. The gradual segregation of poly-disperse mixtures of granular matter during shear is especially dangerous in the pharmaceutical industries, where active ingredients for drugs need to be uniformly mixed in powder form.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	One series of experiments has been aimed at recreating long runout rock avalanches in the laboratory. Such avalanches are extremely rare, but highly destructive events where a large part of a mountain slides very far into a valley at high speed. The distances traveled by such long runout avalanches far exceed what one might expect due to friction between rocks. Modeling such events in the lab should help in risk assessment and mitigation of such destructive events.
Task Progress:	This project has so far resulted in 8 publications, most in flagship journals (2 Phyical. Review Letters publications, 2 Europhysics Letters publications, 2 Physical Review E publicaitons, 1 Journal of Geophysical Research paper, and paper accepted in the International Journal of Earth Sciences). Our main results are described below: (1) Flow instabilities in simple shear flow: We have carried out detailed studies of breaking and reforming of a contact network though shear forces: If granular material starts to flow in a new direction, its old network of particle contacts breaks and a new one must be built. In dry granular matter in a Couette geometry with a free surface we found that destruction of the old network is indicated by compaction, which takes place over a shear length of 0.5 particle diameters, accompanied by a transiently smaller shear strength, which persists until a new contact network develops after ~3-5 particle diameters of shear. Transiently, we thus have a weaker, yet more compact particle arrangement. Our work indicates that simple shear reversal experiments can provide insights into the particle contact network by measuring how such a network forms and breaks. We are currently working on 3D imaging of granular flow during shear reversal (Toiya, Stambaugh and Losert, PRL, 2004). (2) Flow instabilities in avalanches: We studied long runout avalanches in a model hill geometry where avalanches were triggered through horizontal vibration. We found that long runout avalanches, extremely rare but extremely destructive events cannot simply be driven by horizontal shaking for a large range of parameters. Though horizontal shaking fluidizes granular material, it does not provide material with the horizontal momentum necessary to flow long distances after reaching the bottom of a hill. Our work nindicates that any curve in the hillside may drive fluidization of the whole granular layer and strongly increase dissipation (S.J. Friedmann, N. Taberlet, W. Losert, 2004). In related studies we found that for a granular avalanche, the sidewalls are more important than previously thought. In collaboration with the Univ. of Rennes, we found that the dynamic angle of repose of a flowing granular layer is steeper than a static granular heap in a confined geometry. Whether an avalanche flow channel is wide is determined by the ratio of channel width to flow depth, not to particle size as had been previously assumed (Taberlet, Richard, Valance, Losert, Pasini, Jenkins and Delannay, PRL 2003). (3) Segregation: We study segregation in mixtures with more than two sizes of granular matter rotated in a long horizontal drum. Mixtures of particles in a half filled rotated drum segregate into axial bands within bands of increasingly smaller size. Above a threshold number of bands, no axial banding occurs. We found that the dynamics are generic for mixtures of particle types, but that individual particle imaging may be needed to determine the actual mechanism (Newey, Van der Meer, Ozik, Ott and Losert, Europhys. Lett, 2004) We have also started Molecular Dynamics simulations - in collaboration with the University of Rennes. Initial results show the importance of the 3D structure of the segregation pattern in determining the final segregated state (Taberlet, Losert, and Richard, Europhys Lett, 2004). (4) In addition, we have studied particle dynamics in a system where additional attractive forces are present between particles. This work benefited from supplies funded through this grant, though the personnel was funded through a fellowship. Magnetized granular materials self assemble into strings and clusters, depending on the shape of the magnetic near field. In mixtures, particles segregate by magnetic field strength and near field shape (Stambaugh, Lathrop, Ott and Losert 2003; Stambaugh, Smith, Ott and Losert 2004). This work has evolved toward biological applications and the student working on this project (Stambaugh) has graduated.
Bibliography Type:	Description: (Last Updated: 04/11/2007)
Articles in Peer-reviewed Journals	Taberlet, N.; Losert, W.; Richard, P. "Understanding the dynamics of segregation bands of simulated granular material in a rotating drum" Europhys. Lett. 68, 522 , Nov-2004
Articles in Peer-reviewed Journals	Toiya, M.; Stambaugh, J. ; Losert, W. "Transient and oscillatory granular shear flow" Phys Rev Lett. 83, 088001-1 , Aug-2004
Articles in Peer-reviewed Journals	Stambaugh, J.; Smith, Z.; Ott, E.; Losert, W. "Segregation in a monolayer of magnetic spheres" Phys. Rev. E 70, 031304 , Sep-2004
Articles in Peer-reviewed Journals	Newey, M.; Van der Meer, S.; Ozik, J.; Ott, E.; Losert, W. "Band-in-band segregation of multidisperse granular mixtures" Europhys. Lett. 66,205 (2004). , Apr-2004
Awards	K. Newhall "Undergraduate presentation award IREAP- Univ of Maryland" Aug-2004

Awards	Toiya, M. "Best Poster Award, Fluid Dynamics Meeting Univ of Maryland" Nov-2004
Presentation	Losert, W. "Memory in a Sandpile" Statistical Physics Seminar, Univ of Maryland Oct-2004
Presentation	Losert, W. "Memory in a Sandpile" Applied Math Colloquium , New Jersey Institute of Technology Nov-2004
Presentation	Losert, W. "dynamics experiments: Visual demonstrations of non-equilibrium statistical mechanics" Gordon conference on Physics Research and Education Jul-2004
Presentation	Losert, W. "Granular Shear flow: Transients and Segregation Dynamics" LASSP Seminar, Cornell University Apr-2004
Presentation	Losert, W. "The start of granular shear flow: Memory and aging effects" Condensed Matter Physics Seminar, University of Rennes Jan-2004